MANIPAL INSTITUTE OF TECHNOLOGY

Manipal Academy of Higher Education, Manipal

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

THIRD SEMESTER B.TECH. END SEMESTER EXAMINATION - 2025

CS 2104: DATA COMMUNICATION AND COMPUTER NETWORK

Branch: Computer Science & Engineering (CSE) | Credit: 4

Time: 3 Hours | Max. Marks: 50

Instructions: Answer ALL questions. Each question carries designated marks.

PART A

1. (a) Explain OSI 7-layer architecture vs TCP/IP protocol suite with PDU encapsulation. [5 Marks]

1. (b) Compute CRC-16 checksum for given bit stream and explain polynomial error detection. [5 Marks]

PART B

2. (a) Analyze Go-Back-N and Selective Repeat ARQ sliding window protocols with sequence numbers. [5 Marks]

2. (b) Explain IPv4 classless addressing, subnetting, CIDR prefix matching, and NAT traversal. [5 Marks]

PART C

3. (a) Compare Distance Vector Routing (Bellman-Ford count-to-infinity) and Link State (Dijkstra). [10 Marks]

PART D

4. (a) Explain TCP three-way handshake, connection teardown, and AIMD congestion control. [10 Marks]

[END OF QUESTION PAPER - MIT MANIPAL CSE 3RD SEMESTER ARCHIVE]